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TITLE PAGE

TITLE: EFFECTS OF A CONTRAST TRAINING PROGRAM WITHOUT EXTERNAL LOAD ON VERTICAL JUMP, KICKING SPEED, SPRINT AND AGILITY OF YOUNG SOCCER PLAYERS

RUNNING HEAD: CONTRAST TRAINING IN YOUNG SOCCER PLAYERS

AUTHORS: Felipe García-Pinillos[1], Antonio Martínez-Amat[2], Fidel Hita-Contreras[2], Emilio J Martínez-López[1], Pedro A Latorre-Román[1]

[1]University of Jaen, Department of Didactics of Corporal Expression.
[2]University of Jaen, Department of Health Sciences

CORRESPONDING AUTHOR: Felipe García-Pinillos, fegarpi@gmail.com, +34 660 062066, University of Jaen, Campus de Las Lagunillas s/n. D2 Building, Dep. 142. 23071, Jaén (Spain)

The purpose of this study was to determine the effects of a twelve-week contrast training (CT) program (isometric + plyometric), with no external loads, on the vertical jump, kicking speed, sprinting, and agility skills of young soccer players. Thirty young soccer players (age: 15.9 ± 1.43 years; weight: 65.4 ± 10.84 kg; height: 171.0 ± 0.06 cm) were randomized in a control group (n = 13) and an experimental group (n = 17). The CT program was included in the experimental group's training sessions, who undertook it twice a week as a part of their usual weekly training regime. This program included three exercises: one isometric and two plyometric, without external loads. These exercises progressed in volume throughout the training program. Performance in countermovement jump (CMJ), Balsom agility test (BAT), 5-, 10-, 20- and 30-meter sprint, and soccer kick were assessed before and after the training program. A two-factor (group and time) analysis of variance revealed significant improvements ($p < 0.001$) in CMJ, BAT, and kicking speed in the experimental group players. Control group remained unchanged in these variables. Both groups significantly reduced sprint times over 5, 10, 20 and 30 meters ($p < 0.05$). A significant correlation ($r = 0.492$, $p < 0.001$) was revealed between ΔBAT and ΔAverage-Kicking-Speed. Results suggest that a specific CT program without external loads is effective for improving soccer specific skills such as vertical jump, sprint, agility, and kicking speed in young soccer players.

Key words: strength training, youth, post-activation potentiation, specific soccer skills.

INTRODUCTION

Strength training for adolescents can lead to functional (i.e. muscular strength, endurance, power, balance, and coordination) and health benefits (5, 40). Previous research concerning the impact of strength training in the performance of different sport skills has shown increases in performance (13, 15, 23). The most interesting events during a soccer match are represented by high-intensity work, such as sprint, turns, jumps, shots, or tackles (18). The basic movement patterns in soccer require rapid force development and high-power output, as well as the ability to efficiently utilize the stretch-shortening cycle in ballistic movements (12, 31, 37). Along these lines, Cometti (7) stated that strength-training programs must assure transference between the acquired strength and the main technical skills.

Plyometric training (jumping, bounding, and hopping exercises that use the stretch-shortening cycle of the muscle unit) have consistently been proven to improve the production of muscle force and power (16, 39). In particular, the fast force production of the trained muscle profits from such exercises, and smaller increases also become apparent in maximum isometric force (16). Plyometric training has been applied in numerous studies, and there is a general consensus that these physiological adaptations improve soccer specific skills such as agility (25, 28, 37, 38), sprint performance (10, 22, 23, 39), kicking speed (30, 34), and vertical jump performance, all common measures of muscle power (9, 14, 25, 39).

The post-activation potentiation (PAP), as defined by Robbins (32), is a phenomenon by which the exerted muscle force is increased due to its previous contraction. Contrast training (CT) consists in the use of high and low loads in the same strength training session (7, 35). The loads used in CT can engage different regimens of contraction (7). This method is considered very efficient to increase power. In fact, several power training methods have been used extensively with these high- and low-load intensity combinations (2, 3, 20). Therefore, CT is a strength training method supported by the assumption of a PAP of the neuromuscular system (11, 23, 32). It has been postulated that CT provides broad neuromuscular adaptations resulting in greater transfer to a wide variety of performance variables (1, 17). Although CT methods involving heavy loads (>80% 1RM) in conjunction with lighter loads performed ballistically have been reported to improve power (17), some authors have also looked into the acute effects of CT with lighter loads (3, 26, 35). In this sense, Cometti (7) introduced specific guidelines for strength training based on CT. This method involved undertaking, without external loads, an isometric exercise for 40-80 seconds, with six repetitions of the plyometric exercise.

To the extent of our knowledge, this is the first study to analyze the effects of a twelve-week CT program combining isometric and plyometric regimens of contraction without external load for young soccer players. We have found previous studies with similar characteristics but are focused on short-term effects: four-week (1, 19), six-week (23, 36-38), or nine-week protocols (21). Therefore, it is still to be established whether chronic improvements can be accomplished with lighter CT loads over a longer training period.

Our starting point was the fact that combined training provides broader neuromuscular adaptations, which result in greater transfer to a wide variety of performance variables (1, 11, 19, 23, 24, 34, 36-38). We then looked into the benefits of power and fast force training, as well as into the risks that working out with external loads poses for the correct development of young soccer players (5, 18, 40). We hypothesized that a CT program without external loads would improve power and agility scores in young soccer players. As a general goal, this research analyzed the effectiveness of CT (isometric + plyometric) on the vertical jumping, kicking speed, sprinting, and agility of young soccer players.

METHODS

Experimental Approach to the Problem

In this study, we aimed to identify the effects of a twelve-week CT program on the jumping, sprinting, kicking speed, and agility abilities of young soccer players. Additionally, all groups performed their normal soccer training. Using a randomized, between-group design (experimental and control group, EG and CG respectively), 30 soccer players were assessed.

Subjects

Thirty male subjects from a semi-professional soccer academy (age: 15.9 ± 1.43 years; weight: 65.4 ± 10.84 kg; and height: 171.0 ± 0.06 cm) successfully completed the study. Sample size was determined considering previous research on young soccer players of a similar level (1, 23, 34, 37). All players and coaches were informed of the protocol and the experimental risks, and signed an informed consent document before the investigation. Parental consent was obtained for participants under 18 years old. The study was conducted in adherence to the standards of the Declaration of Helsinki (2008 version) and following the European Community's guidelines for Good Clinical Practice (111/3976/88 of July 1990), as well as the Spanish legal framework for clinical research on humans (Real Decreto 561/1993 on clinical trials). The informed consent and the study were approved by the Bioethics

Committee from the University of Jaén, Spain. The study was conducted in-season, when participants attended soccer training three times per week and played competitive matches at least once a week. All participants had been involved in soccer training with this regularity for at least four years prior to the study. Participants were randomly assigned to the experimental group (EG, n = 17) or the control group (CG, n = 13) (Table 1) and the differences in numbers are due to injuries (one participant) and absences in posttest (three participants).

TABLE 1 ABOUT HERE

Procedures

The participants were always kept under surveillance by professional technicians with long experience in strength training. In the countermovement jump (CMJ) the subjects performed a maximal vertical jump starting from a standing position, with arm swing not allowed. All jumps were performed on the FreePower Jump Sensorize (Biocorp, Italy), which provides the following parameters: maximum height of jump (m), peak force ($N \cdot kg^{-1}$), peak power ($W \cdot kg^{-1}$), eccentric work ($J \cdot kg^{-1}$), and concentric work ($J \cdot kg^{-1}$). Subjects performed three trials, with a 30-second recovery period between them. The best result of the three trials was used in the analysis.

Sprint evaluation was accomplished through a speed test that was carried out in a straight 30-meter line (23, 35). Markers were set up at 5 (S5m), 10 (S10m), 20 (S20m) and 30 meters (S30m). Sprint times (s) were measured through 2D photogrammetry. A lateral view of the 30-meter sprint was obtained for all trials using a Casio Exilim EXZR-10 high speed camera (Dover, NJ, USA) with a sampling frequency of 240 Hz. The video camera was placed at a right angle to the running course, 15 meters away, so that a sagittal image of the entire run could be obtained. Video data were digitized using VideoSpeed (vs.1.38, ErgoSport, Granada, Spain).

Agility was evaluated through the Balsom agility test (BAT; 4). This test evaluates the capacity of subjects to quickly change direction. For sprint and agility tests, players were allowed two trials, with a three-minute recovery period in between. The best trial was used for the subsequent analysis in both BAT (4) and sprint test (37). Times (s) were analyzed through 2D photogrammetry, in an identical way to the sprint evaluation.

Soccer kick performance, in terms of ball speed, was measured during shooting. Markers were set up at one and two meters from the initial position of the ball. As in the sprint and BAT time evaluations, the kicking speed, expressed in $m \cdot s^{-1}$, was measured, with the same high speed camera but at a sampling frequency of 480 Hz for this test, and was analyzed through 2D photogrammetry. For this measurement, a ball of standard size and proper pressure according to the rules of Federation Internationale de Football Association (FIFA) was used. Each participant performed three trials with each leg. The best result and the average of both legs were used for statistical analysis (34). The resting period between trials was 40 seconds long. To standardize, we used a two-step run-up. Participants were asked to kick the ball as fast as possible toward the goal, using the instep of the dominant and the non-dominant leg alternatively. They were instructed not to decrease the speed to improve the accuracy of the shot.

A portable eight-polar tactile-electrode impedanciometer (InBody R20, Biospace, Gateshead, UK) was used to measure weight (kg), fat mass (%) and skeletal muscle mass (kg). BMI was calculated as weight (in kilograms) divided by height squared (in meters). Height (cm), was measured with a stadiometer (Seca 22, Hamburg, Germany).

The CT program was performed twice a week for twelve weeks. Participants in the EG performed exercises that consisted of an isometric half squat, with a 90º knee bend, back against the wall, combined with plyometric exercises. The plyometric exercises performed in the CT program were: jumping from the seated position, and single-leg jumping using arm-swing, alternating right and left leg. The program was incorporated into their usual weekly training regime. Participants also continued their usual competitive program of matches.

All EG participants performed a two-week adaptation strength-training program with two sessions. The aims of this training were to optimize the exercise execution, to prevent possible injuries, and to attenuate the learning effect. After finishing this adaptation period, all soccer players were subject to a first evaluation in the following tests: CMJ, kicking speed, S5m, S10m, S20m and S30m, and BAT. After this evaluation, subjects were divided into two groups (EG and CG). The CG performed only the normal soccer training. The CT was performed at the beginning of soccer practice (after warm-up). In each contrast exercise, four to six sets were carried out. Each set was composed of two (weeks first to sixth) to four (weeks sixth to twelfth) exercises, alternating isometric and plyometric exercises, in this order. For isometric exercises, participants performed for 40-80 seconds, while for plyometric exercises, they performed six repetitions. Participants were instructed to maximize jump height. These instructions were emphasized during every session through the use of demonstrations and verbal cues. All training sessions began with a warm-up, consisting of five minutes of low-intensity running, and five minutes of general exercises (high skippings, leg flexions, lateral running, front and behind arm rotation, and sprints). Progression and order of exercises are reported in Table 2.

TABLE 2 ABOUT HERE

Statistical analysis

Descriptive statistics are represented as mean (SD). Tests of normal distribution and homogeneity (Kolmogorov-Smirnov and Levene's) were conducted on all data before analysis. An independent *t* and chi-squared test were used to compare demographic and body composition variables between groups. ANCOVA was performed between groups in pre-test, post-test and post-pre difference, using age as a covariate. We used a two factor (group and time) analysis of variance with repeated measures to assess the training effects on the outcome variables (kicking speed, agility, vertical jump, and sprint ability). Pearson's correlation was executed between the variables analyzed and linear regression between BAT and the average kicking speed difference between post-and pre-training. The reliabilities of sprint ability (S30m), vertical jump (CMJ), agility (BAT) and soccer kick performance (kicking speed in terms of ball speed) were assessed using intraclass correlation coefficients (ICCs) between test-retest and confidence interval (CI). The level of significance was $p < 0.05$. Data analysis was performed using SPSS (version 21, SPSS Inc., Chicago, Ill).

RESULTS

Test-retest reliability analysis of all physical performance tests in the present study shows an ICC of 0.986 (CI 95%=0.972-0.993) for the CMJ; 0.963 (CI 95%=0.927-0.981) for the S30m; 0.883 (CI 95%=0.637-0.963) for the BAT; and 0.974 (CI 95%=0.914-0.992) for the kicking speed.

The results of the tests before and after the CT program are outlined in Table 3. After twelve weeks of CT, significant differences were observed between the EG and the CG. CMJ performance increased for the EG (7.14%, $p<0.001$) while for the CG no significant changes occurred (+2.22%, $p>0.05$). Changes in the mechanical parameters of CMJ also provide interesting information about the effect of the CT program, showing a positive response to training: peak power significantly increased for the EG ($p<0.05$), and, although no significant differences were observed, a clear improvement in peak force and eccentric and concentric work was evident for the EG. The same is not true of the CG, whose values remained unchanged.

TABLE 3 ABOUT HERE

BAT time was reduced by 5.13% for the EG ($p<0.001$). Also, the EG experienced a significant increase ($p<0.001$) in the kicking speed of the dominant (7.12%, $p<0.001$), the non-dominant leg (12.80%, $p<0.001$), and the average of both (9.62%, $p<0.001$). In all these parameters, CG remained unchanged ($p>0.05$): BAT (-0.34%), kicking speed of the dominant leg (-0.97%), the non-dominant leg (0.11%) and the average of both (-0.53%). In addition, after twelve weeks of CT, a reduction was observed on sprint times over 5 m (EG 14.97% and CG 11.44%, $p<0.001$), 10 m (EG 13.36%, $p<0.001$; CG 7.43%, $p<0.01$), 20 m (EG 8.09%, $p<0.001$; CG 5.61%, $p<0.01$), and 30 m (EG 6.26%, $p<0.001$; CG 3.83%, $p<0.01$) and, although results were significant for both groups, the degree of improvement was greater for the EG than the CG.

Table 4 shows results for body composition before and after twelve weeks of CT. No significant differences were found ($p \geq 0.05$) between groups (EG-CG), or between assessment (pre-post).

TABLE 4 ABOUT HERE

Pearson's correlation results between the increase (difference post-pre, Δ) of analyzed variables showed that there were significant correlations to note. A positive correlation was determined between power and explosive strength variables: ΔCMJ and ΔS10m ($r= -0.457$, $p=0.011$), and also between ΔPforce and ΔAverageKickingSpeed ($r= 0.375$, $p=0.041$). In addition, a significant correlation was found between ΔBAT and ΔAverageKickingSpeed ($r= -0.702$, $p<0.001$). The results showed a linear regression between ΔBAT and ΔAverageKickingSpeed ($R^2= 0.492$) (Figure 1).

FIGURE 1 ABOUT HERE

DISCUSSION

The results of this study show that a twelve-week CT program positively affects vertical jump, kicking speed, sprint, and agility performance in young soccer players. After the CT program we found that the EG increased their values in CMJ and kicking speed, as well as a decrease in BAT time and sprint times over 5, 10, 20 and 30 meters. As far as we know, this is the first study to analyze the effects of a CT program with combined isometric and plyometric regimens of contraction with no external load for young soccer players. Others authors have applied CT (23, 30) or plyometric training programs (34, 37, 38), with external loads. Furthermore, we have not found any previous study in which a CT program without external loads was maintained for twelve weeks. In similar studies short-term effects have been analyzed over four-week (1, 19), six-week (23, 36-38), or nine-week protocols (21).

Focusing on the improvements obtained, gains in acceleration (-0.25 s in S5m, and -0.33 s in S10m) and sprint speed (-0.31 s in S20m, and -0.32 s in S30m) confirm those found by other authors (21, 23, 33, 34). Maio Alves et al. (23) found improvements in 5 m (-0.1 s) and 15 m sprints (-0.18 s) after six weeks of a CT program with young soccer players. Likewise, Taïana et al. (36) found smaller improvements in 10 m and 30 m sprints, after a training program identical to that used by Maio Alves et al. (23). Compared to previous studies, the magnitude of the change observed in the current study was greater, and this could be due to the brevity of the training period used in the previous studies. In fact, Kotzamanidis et al. (21) identified a reduction of 0.25 seconds, very similar to ours, in the 30 m sprint time soccer players after the application of a CT program during nine weeks. As for Impellizzeri et al. (19), they found no effect for a four-week program on the 10 m and 20 m sprint time of amateur soccer players. As in the case of the present paper, these previous studies involved a two-sessions-per-week training program. The results found by the above-mentioned authors agree with ours in suggesting that CT programs are useful practice to improve speed over distances between 5 m and 30 m, and that program training duration is an influencing factor. In addition, a positive correlation between ΔS10m and ΔCMJ was found, reinforcing these findings and pointing to an improvement in power and explosive force, which implies neurological adaptations to the CT program (6, 8).

The results obtained in the CMJ performance after the CT program are controversial. While these results are consistent with some previous studies (38), they are also contradictory with the data reported by others (23, 36). In our study, CMJ performance increased (+ 0.03 m, 7.14%) after the CT program. Also, Vaczi et al. (38) found significant improvements after a plyometric training program in only six weeks. However, other authors (23, 36) did not find any significant change in CMJ performance after six weeks of a CT program. Like they themselves explained, this might be due to the frequency of the training session being only once a week. The improvement in jump height indicates that adaptations relating to increases in leg power took place. These adaptations are likely to be neural, as these predominate in the early stages of strength and power training (37) and have been shown to be the main adaptation to plyometric exercise. The adaptations mentioned are an increased neural drive to the agonist muscles and changes in the muscle activation strategies (i.e. improved intermuscular coordination), or changes in the mechanical characteristics of the muscle-tendon complex (25, 38). Similarly to previous studies (1, 19, 21, 23, 36, 38) and despite the fact neurophysiological variables were not directly measured in the current study, the changes in CMJ performance and mechanical parameters (such as the significant improvement in peak power during CMJ) found in this research, would support the rationale that these neurophysiological changes together may improve the ability to store and release elastic energy during the stretch-shortening cycle.

Significant improvements were observed in BAT times (-0.63 s, 5.13%) after the CT program. Previous research has shown improvements in agility performance after strength training programs. Thomas et al. (37) observed that six weeks of plyometric training significantly improved agility (9%) in semiprofessional adolescent soccer players. The greatest improvement in agility (10%) was found in soccer-playing children after eight weeks of plyometric training (27). Miller et al. (28) found 5% and 3% improvements in the t-agility and the Illinois agility tests respectively after six weeks of plyometric training. Vaczi et al. (38) found slight but significant improvements both in the t-agility (2.5%) and in the Illinois agility (1.7%) tests. In contrast, after six weeks of a CT program, Maio Alves et al. (23) did not find significant changes in the 505 Agility Test in youth soccer players. The explanation other authors have given for these results involve the fact that CT does not include any exercise in which athletes had to perform changes in direction, breakings, and start movements, as required by agility tests. However, in our study with a twelve-week CT program for subjects with similar characteristics, we found significant improvements (5.13%) in BAT time. It is risky to compare the results obtained in BAT due to the diversity of tests used to assess agility. The magnitude of the improvement in agility, on one hand, may be influenced by the training status or the age of the participants, demonstrating greater agility enhancement in younger individuals versus adults. Other influencing factors can be the type of the agility test and others such as gender, training status, methods of testing, and differences in duration, intensity, and the types of the exercises used in the training program (25). Overall, improvements in agility after plyometric training and CT programs can be attributed to neural adaptation, and more specifically to increased intermuscular coordination (29). Considering the data obtained in the present study, in which a positive correlation was found between post-pre differences of BAT and kicking speed average, we support the previously stated explanation. Moreover, the fact that body composition remained unchanged during the CT program points at muscle hypertrophy not happening, and also at improvements not being strictly due to muscle adaptations, although neuromuscular changes were of the greatest importance.

Sedano-Campos et al. (34) suggested that there may be a positive transfer of the effects of plyometric training on vertical jump ability to soccer kick performance. The results of the present study are in agreement with this statement, as kicking speed increased ($p<0.001$) on both the dominant (+1.57 m/s, 7.12%) and the non-dominant legs (+2.18 m/s, 12.80%), and, consequently, on the average of both (+1.87 m/s, 9.62%) after a twelve-week CT program. However, in male players, the effects of strength training on kicking performance are controversial. Although some studies reported an increase in performance after the application of training programs involving explosive strength (30, 34, 36), maximal strength (36), or mixed technical and strength training (30, 34), others found the opposite to be true (23). These diverging results from previous research could be due to training program characteristics implemented and, more specifically to the program training duration (34). Data reported (23, 34) also revealed that although six weeks of plyometric training was enough time to produce significant improvements in explosive strength, players required twelve weeks to produce significant increases in kicking speed. This could also be due to other influencing factors such as the maximal strength of the muscles involved, the rate of force development, neuromuscular coordination, the linear and angular velocities of ankle in the kicking leg, and the level of coordination between agonist and antagonists (7, 11, 17, 18, 20, 21, 38). Kicking speed is influenced by the particular features of the stretch-shortening cycle in the muscles involved (24). Plyometric exercises induce neuromuscular adaptations to the stretch reflex, which can lead to greater recruitment of motor units during muscular contraction (5, 31) and, as shown by previous studies, to increased performance in ballistic movements (like soccer kick performance) in terms of ball speed (30, 34).

Correlations between parameters of explosive strength and power have been widely studied in soccer players (12, 13, 18). However, the Pearson´s correlation analysis performed in this study allowed us to add to the known correlations a new correlation between ΔBAT and ΔAverageKickingSpeed ($r= -0.702$, $p<0.001$). To our knowledge, this correlation has not previously been reported in the literature, and it points to an association ($R^2= 0.492$) between influential parameters in soccer performance (agility and kicking speed), which should be incorporated to daily workout routines.

To summarize, our findings support the idea that a CT program without external loads is effective for improving power and agility in young soccer players and, consequently, soccer specific skills such as sprinting, vertical jumping, agility and kicking speed. In summary, the present study yielded data about an improvement in CMJ based on higher power, about a reduction in BAT and sprint times, and about gains in kicking speed performance which indicate an increase in power and fast force. Also, the maintenance of body composition parameters showed no muscle hypertrophy. We may therefore conclude that these adaptations could be attributed to neuromuscular changes that led to a better transfer of strength gain to specific skills in young soccer players.

PRACTICAL APPLICATIONS

From a practical point of view, it must be considered that a twelve-week CT program (isometric + plyometric) without external load enhances the physical capacities related to improved performance in soccer. In young soccer players, such capacities include speed, agility, power, and explosive and specific strength, and may have a high degree of transference into game-play performance. Thus, a twice weekly short-term high intensity CT program, implemented as a substitute for some soccer drills within regular in-season soccer practice, can enhance explosive performance in young soccer players compared with soccer training alone. These improvements can be achieved using 48- to 72-hour rest periods between CT sessions. Despite the improvements that this study found after this training method alone (in jump capacity, agility, speed, and kicking speed), we recommend that it be integrated into a comprehensive training program aimed at developing specific and performance-critical technical abilities (particularly for young players). In addition, this training method should be accompanied by a program of physical conditioning in order to optimize training adaptations.

REFERENCES

1. Argus, CK, Gill, ND, Keogh, JWL, McGuigan, MR, and Hopkin, WJ. Effects of two contrast training programs on jump performance in rugby union players during a competition phase. *Int J Sports Physiol Perform* 7: 68-75, 2012.

2. Baker, D. Acute and long-term power responses to power training of an elite power athlete. *Strength Cond* 23: 47–56, 2001.

3. Baker, D. A series of studies on training of high-intensity muscle power in rugby league soccer players. *J Strength Cond Res* 15: 198– 209, 2001.

4. Balsom, PD. *Evaluation of physical performance.* In: B. Ekblom (Ed.). Football (soccer). Blackwell, London 1994; pp. 102-123.

5. Behm, DG, Faigenbaum, AD, Falk, P, and Klentrou, P. Canadian Society for Exercise Physiology position paper: resistance training in children and adolescents**.** *App Physiol Nutr Metab* 33(3): 547-561, 2008.

6. Chelly, MS, Ghenem, MA, Abid, K, Hermassi, S, Tabka, Z, and Shephard, RJ. Effects of in-season short-term plyometric training program on leg power, jump and sprint performance of soccer players. *J Strength Cond Res* 24(10):2670-6, 2010.

7. Cometti, G. *Los Métodos modernos de Musculación.* Editorial Paidotribo, Barcelona, Spain: 1998.

8. Comfort, P, Stewart, A, Bloom, L, and Clarkson, B. Relationships between strength, sprint, and jump performance in well-trained youth soccer players. *J Strength Cond Res* 28(1): 173-177, 2014.

9. Clutch, D, Wilton, M, McGown, C, and Bryce, GR. The effect of depth jumps and weight training on leg strength and vertical jump. *Res Q* 54: 5–10, 1983.

10. Delecluse, C, Van Coppenolle, H, Willems, E, Van Leemputte, M, Diels, R, and Goris, M. Influence of high-resistance and high-velocity training on sprint performance. *Med Sci Sports Exerc* 27: 1203–1209, 1995.

11. Docherty, D, Robbins, D, and Hodgson, M. Complex training revisited: A review of its current status as a viable training approach. *Strength Cond* 26: 52–57, 2004.

12. Ellis, L, Gastin, P, Lawrence, S, Savage, B, Buckeridge, A, Stapff, A, Tumilty, D, Quinn, A, Woolford, S, and Young, W. Protocols for the physiological assessment of team sports players. In: *Physiological Tests for Elite Athletes.* C.J. Gore, ed. Champaign: Human Kinetics, 2000. pp. 128–144.

13. Fatouros, IG, Jamurtas, AZ, Leontsini, D, Taxildaris, K, Aggelousis, N, Kostopoulos, N, and Buckenmeyer, P. Evaluation of plyometric exercise training, weight training and their

combination on vertical jumping performance and leg strength. *J Strength Cond Res* 14: 470–476, 2000.

14. Gehri, DJ, Ricard, MD, Kleiner, DM, and Kirkandall, TD. A comparison of plyometric training techiques for improving vertical jump ability and energy production. *J Strength Cond Res* 12: 85–89, 1998.

15. Gourgoulis, V, Aggeloussis, N, Kasimatis, P, Mavromatis, G, and Garas, A. Effect of submaximal half squats warm-up program on vertical jump ability. *J Strength Cond Res* 17: 342–344, 2003.

16. Hakkinen, K, Alen, M, and Komi, PV. Changes in isometric force- and relaxation-time, electromyographic and muscle fibre characteristics of human skeletal muscle during strength training and detraining. *Acta Physiol Scand* 125: 573–585, 1985.

17. Harris, GR, Stone, MH, O'Bryant, HS, Proulx, CM, and Johnson RL. Short-term performance effects on high power, high force, or combined weight-training methods. *J Strength Cond Res* 14: 14–20, 2000.

18. Hoff, J, and Helgerud J. Endurance and Strength Training for Soccer Players. Physiological Considerations. *Sports Med* 34 (3): 165-180, 2004.

19. Impellizzeri, FM, Rampinini, E, Castagna, C, Martino, F, Fiorini, S, and Wisloff U. Effects of plyometric training on sand versus grass on muscle soreness and jumping and sprinting ability in soccer players. *Br J Sports Med* 42: 42-46, 2008.

20. Kawamori, N and Haff, GG. The optimal training load for the development of muscular power. *J Strength Cond Res* 18: 675–684, 2004.

21. Kotzamanidis, C, Chatzopoulos, D, Michailidis, C, Papaiakovou, G, and Patikas, D. The effect of combined high-intensity strength and speed training program on the running and jumping ability of soccer players. *J Strength Cond Res* 19: 369–375, 2005.

22. Kraemer, WJ, Ratamess, NA, Volek, JS, Mazzetti, SA, and Gomez, AL. The effect of the meridian shoe on vertical jump and sprint performances following short-term combined plyometric/sprint and resistance training. *J Strength Cond Res* 14: 228–238, 2000.

23. Maio Alves, JM, Rebelo, AN, Abrantes, C, and Sampaio, J. Short-term effects of complex and contrast training in soccer player´s vertical jump, sprint and agility abilities. *J Strength Cond Res* 24, 4: 936-946, 2010.

24. Manolopoulos, E, Papadopoulos, C, and Kellis, E. Effects of combined strength and kick coordination training on soccer kick biomechanics in amateur players. *Scan J Med Sci Sport* 16: 102–110, 2006.

25. Markovic, G, and Mikulic, P. Neuro-musculoskeletal and performance adaptations to lower-extremity plyometric training. *Sports Med* 40: 859-895, 2010.

26. McBride, JM, Triplett-McBride, T, Davie, A, and Newton, RU. The effect of heavy- vs. light-load jump squats on the development of strength, power, and speed. *J Strength Cond Res* 16:75–82, 2002.

27. Meylan, C, and Malatesta, D. Effects of in-season plyometric training within soccer practice on explosive actions of young players. *J Strength Cond Res* 23: 2605-2613, 2009.

28. Miller, MG, Herniman, JJ, Richard, MD, Cheatham, CC, and Michael, TJ. The effects of a 6-week plyometric training program on agility. *J Sports Sci Med* 5: 459-465, 2006.

29. Myer, GD, Ford, KR, Brent, JL, and Hewett TE. The effects of plyometric vs. dynamic stabilization and balance training on power, balance, and landing force in female athletes. *J Strength Cond Res* 20: 345-53, 2006.

30. Perez-Gomez, J, Olmedillas, H, Delgado-Guerra, S, Ara, I, Vicente-Rodriguez, G, Ortiz RA, Chavarren, J, and Calbet, JA. Effects of weight lifting training combined with plyometric exercises on physical fitness, body composition, and knee extension velocity during kicking in football. *Appl Physiol Nutr Metab* 33(3): 501-10, 2008.

31. Plisk, SS. Speed, agility and speed endurance development. In: *Essentials of Strength Training and Conditioning (2nd ed.)*. Baechle T.R. and Earle, R.W. eds. Human Kinetics, 2000. pp. 427–470.

32. Robbins, DW. Postactivation potentiation and its practical applicability: A brief review. *J Strength Cond Res* 19: 453–458, 2005.

33. Ronnestad, BR, Kvamme, NH, Sunde, A, and Raastad T. Short-term effects of strength and plyometric training on sprint and jump performance in professional soccer players. *J Strength Cond Res* 22: 773-780, 2008.

34. Sedano Campo, S, Vaeyens, R, Philippaerts, RM, Redondo, JC, De Benito, AM, and Cuadrado, C. Effects of lower-limb plyometric training on body composition, explosive strength, and kicking speed in female soccer players. *J Strength Cond Res* 23(6): 1714-1722, 2009.

35. Smilios, I, Pilianidis, T, Sotiropoulos, K, Antonakis, M, and Tokmakidis, SP. Short-term effects of selected exercise and load in contrast training on vertical jump performance. *J Strength Cond Res* 19: 135–139, 2005.

36. Taıäna, F, Gréhaigne, J, and Cometti, G. The influence of maximal strength training of lower limbs of soccer players on their physical and kick performances. Science and Football II. In: *Proceedings of the Second World Congress of Science and Football*. Reilly, T, Clarys, J, and Stibbe, A, eds. Taylor and Francis, Eindhoven, the Netherlands: 1991.

37. Thomas, K, French, D, and Hayes, PR. The effect of two plyometric training techniques on muscular power and agility in youth soccer players. *J Strength Cond Res* 23(1): 332–335, 2009.

38. Váczi, M, Tollár, J, Meszler, B, Ivett Juhász, I, and Karsai, I . Short-term high intensity plyometric training program improves strength, power and agility in male soccer players. *J Hum Kinetics* 36: 17-26, 2013.

39. Wagner, DR and Kocak, MS. A multivariate approach to assessing anaerobic power following a plyometric training program. J Strength Cond Res 11: 251–255, 1997.

40. Warren, K, Young, J, and Metzl, D. Strength Training for the Young Athlete. *Pediatr Ann* 39 (5), 293 – 299, 2010.

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TABLES LEGENDS

Table 1. Characteristics of young soccer players (mean ± SD).

Table 2. Detailed description of the twelve-week CT program, including exercises, number of sets and repetitions, and recovery and work times (s).

Table 3. Results (mean [SD]) of tests performed by soccer players before and after the CT program.

Table 4. Results obtained from the body composition assessment (mean ± SD), before and after the CT program in CG and EG.

FIGURE LEGENDS

Figure 1. Regression graph between increase in performance (Δpost-pre) of average kicking speed (m/s) and BAT (s).

Table 1. Characteristics of young soccer players (Mean ± SD)

		Control group (n=13)	Experimental group (n=17)	p
Age (y)		16.38 ± 1.5	15.47 ± 1.28	0.083
Height (cm)		169 ± 0.06	172 ± 0.05	0.193
Weight (kg)		61.53 ± 9.46	68.32 ± 11.18	0.089
BMI*		21.42 ± 1.86	23.03 ± 2.56	0.067
Dominant leg n (%)	Right	12 (92.31)	14 (82.35%)	0.098
	Left	1 (7.69)	3 (17.65%)	
Injuries* * n (%)	Yes	3 (21.4)	5 (29.4)	0.613
	No	11 (78.6)	12 (70.6)	

*BMI: Body Mass Index; **Injuries suffered in the last season.

Table 2. Detailed description of the twelve-week CT program, including exercises, number of sets and repetitions, and recovery and work times (s).

Weeks (1-12)	Combination of exercises	Methodology (sets x recovery)
Weeks 1-2	Ex.1 (40 s) + Ex.2 (6 reps)	4 x 2 min
Weeks 3-4	Ex.1 (60 s) + Ex.3 (6 reps)	5 x 2 min
Weeks 5-6	Ex.1 (80 s) + Ex.2 or Ex.3 alternatively (6 sets x 6 reps)	6 x 2 min
Weeks 7-8	Ex.1 (40 s) + Ex.3 (6 reps) + Ex.1 (40 s) + Ex.2 (6 reps)	4 x 2 min
Weeks 9-10	Ex.1 (40 s) + Ex.3 (6 reps) + Ex.1 (40 s) + Ex.2 (6 reps)	5 x 2 min
Weeks 11-12	Identical to weeks 9-10	6 x 2 min

Reps.: repetitions; min: minutes; s: seconds; Ex.1: Exercise 1 (90º isometric half squat exercise); Ex.2: Exercise 2 (Jump from the seated position); Ex.3: Exercise 3 (Single-leg jump, alternating right and left).

Table 3. Results (mean [SD]) of tests performed by soccer players before and after the CT program.

Performance variables	Control Group (n=13)						Experimental Group (n=17)					
	Pretest Mean (SD)	CV (%)	Posttest Mean (SD)	CV (%)	Δ%	p	Pretest Mean (SD)	CV (%)	Posttest Mean (SD)	CV (%)	Δ%	p
CMJ (m)	0.45 (0.03)	6.66	0.46 (0.03)	6.52	2.22	0.058	0.42 (0.06)	14.28	0.45 (0.04)	8.88	7.14	<0.001
Peak force (N.kg^{-1})	16.72 (3.82)	22.84	15.54 (2.96)	19.04	-7.04	0.153	16.11 (3.29)	20.42	17.00 (2.89)	17	5.56	0.212
Peak power (W.kg^{-1})	31.55 (5.23)	16.57	32.18 (5.17)	16.06	2.00	0.642	29.07 (6.80)	23.39	31.54 (6.57)	20.83	8.50	0.044
Eccentric work (J.kg^{-1})	-2.51 (0.73)	29.08	-2.56 (0.44)	17.18	-1.99	0.768	-2.43 (0.85)	34.97	-2.68 (1.02)	38.05	-10.37	0.097
Concentric work (J.kg^{-1})	6.89 (0.80)	11.61	7.07(0.65)	9.19	2.61	0.838	5.71 (3.56)	62.34	7.12 (1.39)	19.52	24.72	0.074
S5m (s)	1.66 (0.17)	10.24	1.46 (0.18)	12.32	-11.44	<0.001	1.67 (0.23)	13.77	1.42 (0.18)	12.67	-14.97	<0.001
S10m (s)	2.42 (0.15)	6.19	2.24 (0.17)	7.58	-7.43	0.008	2.47 (0.25)	10.12	2.14 (0.30)	14.01	-13.36	<0.001
S20m (s)	3.74 (0.17)	4.54	3.53 (0.21)	5.94	-5.61	0.001	3.83 (0.29)	7.57	3.51 (0.24)	6.83	-8.09^	<0.001
S30m (s)	4.96 (0.18)	3.62	4.77 (0.24)	5.03	-3.83	0.001	5.11 (0.35)	6.84	4.79 (0.30)	6.26	-6.26^	<0.001
BAT (s)	11.93 (0.58)	4.86	11.89 (0.46)	3.77	-0.34	0.494	12.29 (0.54)	4.39	11.66 (0.43) #	3.68	-5.13^^	<0.001
Kicking speed dominant leg (m/s)	23.57 (1.51)	6.40	23.34 (1.42)	6.08	-0.97	0.144	22.03 (1.50)	6.80	23.60 (1.72)	7.28	7.12^^	<0.001
Kicking speed non-dominant leg (m/s)	17.79 (2.30)	12.92	17.81 (2.07)	11.62	0.11	0.925	17.03(1.70)	9.98	19.21 (2.04) ##	10.61	12.80^^	<0.001
Kicking Speed average (m/s)	20.68(1.48)	7.15	20.57 (1.31)	6.36	-0.53	0.433	19.53 (1.41)	7.21	21.41 (1.76) ##	8.22	9.62^^	<0.001

Countermovement Jump (CMJ); Sprint time 5 m (S5m); Sprint time 10 m (S10m); Sprint time 20 m (S20m); Sprint time 30 m (S30m); Balsom Agility Test (BAT); Δ%: difference post-pre, in percentage; SD (Standard deviation); Coefficient of Variation (CV); # p<0.05 and ## p<0.01 denote post differences with control group; ^ p<0.05 and ^^ p<0.001 denote post-pre differences with control group.

Table 4. Results obtained from the body composition assessment (mean ± SD), before and after the CT program in CG and EG.

Performance variables	Control Group (n=7)						Experimental Group (n=13)					
	Pretest Mean (SD)	CV (%)	Posttest Mean (SD)	CV (%)	Δ%	p	Pretest Mean (SD)	CV (%)	Posttest Mean (SD)	CV (%)	Δ%	p
BMI	21.42 (1.86)	8.68	21.62 (1.87)	8.64	0.94	0.065	23.03 (2.55) #	11.07	23.04 (2.50)	10.85	0.04	0.915
% Fat mass	12.11 (3.81)	31.46	11.70 (3.83)	32.73	-3.42	0.463	14.44 (6.66)	46.12	14.79 (7.32)	49.49	2.44	0.475
Skeletal muscle mass (kg)	30.45 (4.02)	13.20	31.05 (3.92)	12.62	1.97	0.077	32.96 (4.44)^	13.47	33.01 (3.60)*	10.90	0.16	0.855

Body mass index (BMI); SD (Standard deviation); Coefficient of Variation (CV); Δ%: difference post-pre, in percentage; # $p<0.05$ and ^ $p<0.01$ denote pretest differences with control group; * $p<0.05$ denotes postest differences with control group.

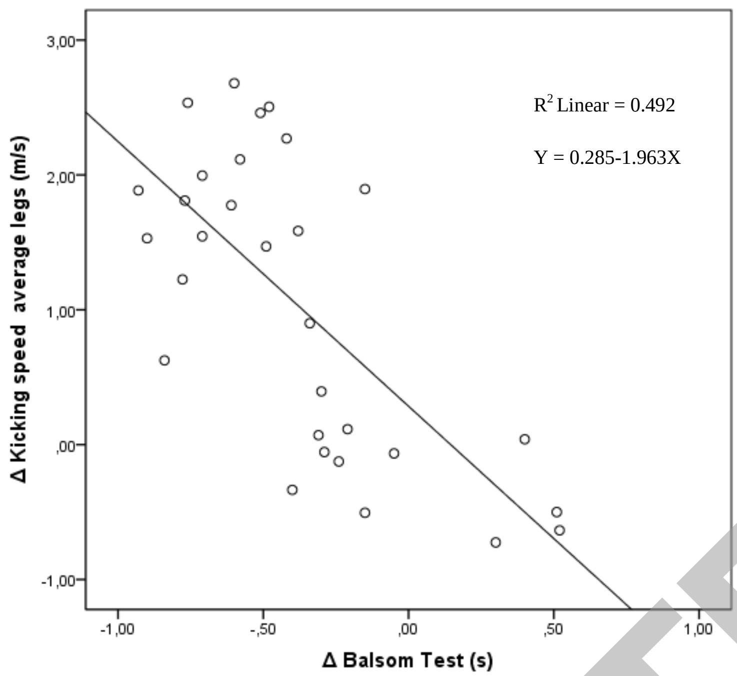

Figure 1. Regression graph between increase in performance (Δpost-pre) of average kicking speed (m/s) and BAT (s).